A comparative study of bulk RNA-seq deconvolution methods using scRNA-seq references

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SUMMARY OF THE FINAL PROJECT

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Abstract

Accurate characterization of the tumor microenvironment (TME) is essential for understanding disease progression and therapeutic response in metastatic prostate adenocarcinoma. While bulk RNA sequencing remains widely used to profile tumor samples, it captures averaged gene expression across heterogeneous cellular populations, limiting direct interpretation of immune cell infiltration. Computational deconvolution methods aim to overcome this limitation by estimating cell-type proportions from bulk transcriptomic data using reference expression profiles derived from single-cell RNA sequencing (scRNA-seq).

This project proposes a comparative evaluation of EPIC, CIBERSORTx, MuSiC, and BisqueRNA applied to bulk RNA-seq data from prostate adenocarcinoma tumors (GSE297742), using scRNA-seq data from metastatic lesions and peripheral blood mononuclear cells (GSE297652) as an external, non sample-matched reference. This cross-cohort setting reflects common real-world scenarios in which scRNA-seq atlases are used to deconvolve independent bulk datasets.

The study aims to assess how preprocessing strategies, gene harmonization, and reference construction influence inferred immune cell proportions within the prostate tumor microenvironment. The central hypothesis is that methodological choices and

reference mismatch significantly impact deconvolution robustness and reproducibility. By systematically benchmarking multiple frameworks in a defined clinical context, this work seeks to provide practical guidance for immune profiling in prostate cancer transcriptomic studies.

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1. Justification of the Proposal

Prostate adenocarcinoma is one of the most prevalent malignancies in men worldwide and remains a major cause of cancer-related mortality, particularly in advanced and metastatic stages. Increasing evidence highlights the importance of the tumor microenvironment (TME), and especially immune cell infiltration, in shaping tumor progression, therapeutic response, and resistance mechanisms. Reliable characterization of immune composition within prostate tumors is therefore essential for translational oncology research.

Bulk RNA sequencing is widely used to profile tumor transcriptomes due to its scalability and availability in public repositories. However, bulk transcriptomic data represent averaged gene expression across heterogeneous cell populations, obscuring direct interpretation of cellular composition. This limitation is particularly relevant in prostate cancer, where immune infiltration patterns can be heterogeneous and relatively subtle compared to more immunogenic tumor types.

Computational deconvolution methods aim to infer cell-type proportions from bulk RNA-seq data using reference expression profiles derived from purified cells or single-cell RNA sequencing (scRNA-seq). In this project, bulk RNA-seq data from prostate adenocarcinoma tumors (GSE297742) and scRNA-seq data derived from metastatic lesions and peripheral blood mononuclear cells (GSE297652) will be used. Importantly, these datasets are not sample-matched, reflecting a common real-world scenario in which scRNA-seq atlases are used as external references to deconvolve independent bulk cohorts.

While previous studies have benchmarked deconvolution tools, many evaluations rely on simulated mixtures, matched datasets, or idealized reference conditions. Fewer studies explicitly address robustness in cross-cohort settings, where differences in preprocessing pipelines, gene harmonization strategies, and reference construction may substantially influence deconvolution outputs. This project directly addresses this methodological gap by systematically comparing EPIC, CIBERSORTx, MuSiC, and BisqueRNA in a defined metastatic prostate cancer context.

From a scientific perspective, this work contributes to improving reproducibility, methodological transparency, and practical guidance in computational immunogenomics. From a broader perspective, although focused on prostate adenocarcinoma, the methodological insights derived from this study may inform immune deconvolution practices in other solid tumors where non sample-matched scRNA-seq references are commonly used.

2. Personal motivation

My motivation to undertake this Final Master's Project arises from the intersection of my academic background, prior research experience, and long-term interest in computational biology. The topic of immune cell deconvolution was initially proposed by my external supervisor. After conducting an independent review of the relevant literature and exploring current methodological developments in the field, I decided to pursue this line of research due to both its scientific relevance and its strong alignment with my professional goals.

During my internship at the Technical University of Munich, I worked on RNA-seq analysis in the context of tumor-educated platelets, applying statistical frameworks such as edgeR and limma to investigate transcriptomic biomarkers. This experience exposed me to the practical challenges of bulk transcriptomic data, including normalization, differential expression analysis, and biological interpretation of heterogeneous signals.

Beyond its academic value, this project also reflects my broader interest in translational bioinformatics and data-driven healthcare innovation. I am motivated by the challenge of developing reproducible and methodologically sound computational workflows that can ultimately enhance biological interpretation and support precision medicine research.

3. Objectives

To comparatively evaluate immune deconvolution methods in the context of metastatic prostate adenocarcinoma, using non sample-matched single-cell RNA-seq references to infer immune cell composition from bulk RNA-seq tumor profiles, with a focus on methodological robustness and reproducibility within the tumor microenvironment.

To achieve the main objective, the project will pursue the following secondary objectives:

- To characterize the immune landscape of metastatic prostate adenocarcinoma bulk RNA-seq samples (GSE297742) through computational deconvolution.

- To construct and harmonize reference expression profiles derived from scRNA-seq data of metastatic prostate lesions and peripheral blood mononuclear cells (GSE297652).

- To systematically compare EPIC, CIBERSORTx, MuSiC, and BisqueRNA under a cross-cohort reference setting.

- To assess the impact of preprocessing decisions, gene intersection strategies, and reference construction on inferred immune cell proportions within the prostate tumor microenvironment.

- To evaluate concordance and variability across deconvolution frameworks in a clinically relevant metastatic tumor context, rather than under simulated or matched conditions.

- To derive methodological recommendations for immune profiling in prostate cancer transcriptomic studies and similar solid tumor settings.

4. Hypothesis

Immune cell deconvolution results derived from bulk RNA-seq data are significantly influenced by preprocessing strategies and reference construction choices in single-cell RNA-seq datasets, leading to measurable variability across computational methods.

To test the main hypothesis, the project will address the following research questions:

1. How do different single-cell RNA-seq preprocessing strategies (e.g., quality control thresholds, normalization approaches, gene filtering) affect the construction of reference signatures?

2. To what extent does gene harmonization between bulk and single-cell datasets influence deconvolution outputs?

3. Do different deconvolution algorithms (e.g., EPIC, CIBERSORTx, MuSiC, BisqueRNA) produce consistent immune cell proportion estimates under standardized preprocessing conditions?

4. How sensitive are deconvolution results to variations in reference construction and gene selection strategies?

5. Can methodological standardization reduce variability across methods and improve reproducibility of inferred immune cell proportions?

5. Impact on Sustainability, Socio-Ethical, and Diversity

This Final Master's Project is framed within the transversal competence of Ethical and Global Commitment (CCEG) promoted by the Universitat Oberta de Catalunya (UOC) (Universitat Oberta de Catalunya, 2022). This competence encourages academic and professional practice that is ethical, socially responsible, sustainable, and respectful of diversity and human rights. The present project integrates these dimensions throughout its conceptual design, methodological framework, and data usage strategy, in alignment with the United Nations Sustainable Development Goals (SDGs) (United Nations, 2015).

5.1 Sustainability

From a sustainability perspective, this project contributes primarily to:

- SDG 3 – Good Health and Well-being
- SDG 9 – Industry, Innovation and Infrastructure

The study focuses on metastatic prostate adenocarcinoma, a clinically significant disease with substantial global health impact. By improving methodological robustness in immune cell deconvolution, the project supports more accurate interpretation of tumor transcriptomic data, which may contribute indirectly to better understanding of tumor–immune interactions and future translational research efforts. In this sense, the project aligns with SDG 3 by promoting advancements in biomedical knowledge that may inform improved diagnostic and therapeutic strategies.

Furthermore, the project contributes to SDG 9 by advancing computational innovation in biomedical data analysis. The development of reproducible and robust computational workflows strengthens research infrastructure and promotes responsible digital transformation in health research.

Importantly, this study exclusively relies on publicly available transcriptomic datasets retrieved from the NCBI Gene Expression Omnibus (GEO). By reusing existing anonymized patient data rather than generating new experimental sequencing data, the project promotes sustainable research practices, minimizing additional resource consumption and laboratory costs while maximizing the scientific value of publicly funded datasets.

5.2 Ethical Behaviour and Social Responsibility

The project adheres strictly to ethical standards in biomedical data analysis. All datasets used (GSE297742 and GSE297652) are publicly available and anonymized, with no personally identifiable information accessible. As such, the study complies with data protection principles consistent with the General Data Protection Regulation (GDPR).

The research does not involve direct patient recruitment, experimental interventions, or access to confidential clinical records. Instead, it relies on secondary analysis of ethically approved and publicly shared transcriptomic data.

Methodological transparency is also an ethical commitment. By clearly documenting preprocessing steps, harmonization strategies, and analytical decisions, the project promotes reproducibility and accountability in computational research, reducing the risk of opaque or irreproducible findings. This aligns directly with the CCEG principles of honesty, responsibility, and professional integrity (Universitat Oberta de Catalunya, 2022).

5.3 Diversity and Human Rights

Biomedical transcriptomic datasets may reflect demographic imbalances, including potential underrepresentation of certain age groups, ethnicities, or clinical subtypes. Although prostate adenocarcinoma is a male-specific disease, variability in ethnicity, disease stage, and treatment background may still influence transcriptomic profiles.

Where metadata are available, the project will acknowledge the demographic composition of the cohorts and explicitly discuss potential biases or limitations. If demographic balance cannot be assessed due to incomplete metadata, this limitation will be transparently reported.

Recognizing these limitations is essential to avoid overgeneralization and to ensure responsible interpretation of computational immune profiling results. This approach aligns with the CCEG dimension related to diversity and human rights and supports inclusive research practices consistent with the SDGs (United Nations, 2015).

6. Methodology

6.1 Research Strategy

This thesis follows a quantitative empirical research strategy within the field of computational bioinformatics. More specifically, it adopts a comparative benchmarking framework aimed at evaluating the robustness of established bulk RNA-seq

deconvolution methods under a realistic translational setting. Rather than proposing a new algorithm, the study focuses on how methodological choices related to reference construction, preprocessing, and harmonization influence the behavior of existing deconvolution approaches.

The study does not generate new biological data. Instead, it performs secondary analysis of publicly available transcriptomic datasets retrieved from the NCBI Gene Expression Omnibus (GEO), namely GSE297742 and GSE297652 (National Center for Biotechnology Information [NCBI], n.d.a, n.d.b), which are associated with the study by Kim et al. (2026). These datasets provide, respectively, bulk RNA-seq data from prostate adenocarcinoma tumor biopsy samples and single-cell RNA-seq data from metastatic lesion biopsies and peripheral blood mononuclear cells (PBMCs) from M1-stage prostate cancer patients.

The bulk and single-cell datasets used in this study are not matched at the individual sample level, since no direct identifier is available to link both modalities one-to-one. However, they do not originate from fully unrelated contexts, as both are associated with the same broader study framework (Kim et al., 2026). This design therefore reflects a partially aligned but non-paired scenario, which is methodologically relevant because it resembles a common practical situation in translational transcriptomics: the use of biologically related single-cell data as a reference source for bulk deconvolution in the absence of direct sample-level pairing.

Accordingly, the main goal of the methodology is not direct biological validation at the patient level, but the assessment of methodological robustness. In this context, robustness refers to the stability and interpretability of deconvolution results under different preprocessing, harmonization, and reference-construction choices.

6.2 Analytical Framework

The analytical pipeline is implemented as a modular and reproducible workflow in R Markdown, with clearly separated stages to ensure transparency, traceability, and methodological consistency. The overall design is informed by methodological best practices in single-cell analysis and by benchmarking studies in transcriptomic deconvolution (Luecken & Theis, 2019; Avila Cobos et al., 2020; Heumos et al., 2023; Xu et al., 2025).

Stage 1: Dataset Acquisition and Initial Assessment

Two publicly available GEO datasets are used in this study: bulk RNA-seq data from prostate adenocarcinoma tumor biopsy samples (GSE297742) and single-cell RNA-seq data from metastatic lesion biopsies and PBMCs from M1-stage prostate cancer patients (GSE297652) (NCBI, n.d.a, n.d.b). Both datasets are linked to the study by Kim et al.

(2026), which serves as the article of origin for the biological and experimental context of the data.

At this stage, sample metadata, expression matrices, and gene annotations are examined in order to assess dataset structure, determine the degree of compatibility between modalities, and define harmonization strategies prior to downstream analyses. This initial assessment is especially important because the datasets are biologically related but not directly paired, which makes transparent documentation of assumptions and constraints essential.

Stage 2: Single-Cell RNA-seq Preprocessing and Annotation

Single-cell preprocessing follows established best-practice recommendations for scRNA-seq analysis (Luecken & Theis, 2019; Heumos et al., 2023). The workflow includes quality control filtering, normalization, dimensionality reduction, clustering, and marker-based interpretation of cell populations. In the current implementation, normalization is performed using SCTransform, as implemented in Seurat, which is widely used for variance stabilization and normalization in single-cell workflows (Hafemeister & Satija, 2019; Stuart et al., 2019).

Cluster annotation is performed using a semi-structured strategy based on canonical immune markers, cluster-level expression scoring, and biologically interpretable marker combinations. This strategy is intended to reduce subjectivity while preserving transparency in how labels are assigned. Cell populations are annotated at the cluster level and categorized according to confidence, allowing uncertain or weakly supported clusters to be excluded from later reference construction. This step is particularly relevant because benchmarking studies have shown that annotation granularity and cell-type definition can substantially influence downstream deconvolution performance (Avila Cobos et al., 2020; Xu et al., 2025).

Stage 3: Reference Signature Construction

A cell-type reference expression matrix is constructed from the annotated single-cell data by aggregating normalized expression profiles within each retained cell type. Only clusters with sufficiently confident annotations are included in the final reference. The resulting matrix is intended to represent the main immune and tumor-associated populations captured by the available scRNA-seq data and is used as the reference input for downstream deconvolution analyses.

This stage is treated as a central methodological step rather than as a purely preparatory one. Recent literature has emphasized that reference construction strongly affects deconvolution accuracy, particularly in settings where cell populations may be incompletely represented or where annotation and granularity decisions influence the

biological meaning of the final signatures (Maden et al., 2023; Ivich et al., 2025; Xu et al., 2025).

Stage 4: Preliminary Technical Validation via Pseudobulk

Before applying deconvolution methods to the bulk RNA-seq cohort, controlled pseudobulk mixtures are generated from the annotated single-cell dataset. These mixtures contain known or traceable cell-type proportions and serve as a preliminary technical validation step.

The purpose of this stage is twofold: first, to verify that each deconvolution framework is implemented correctly under controlled conditions; and second, to assess whether major cell populations can be recovered at a basic level when the composition of the mixture is known. This step is used as a methodological consistency check and should not be interpreted as a substitute for biological validation in real tumor samples. Its use is consistent with current benchmarking practice, where pseudo-bulk mixtures are commonly employed to test deconvolution pipelines under controlled settings (Avila Cobos et al., 2020; White et al., 2024; Xu et al., 2025).

Stage 5: Bulk RNA-seq Harmonization

Bulk RNA-seq data are preprocessed and harmonized with the single-cell reference prior to deconvolution. This includes evaluation of shared gene space, standardization of gene identifiers, and explicit definition of gene intersection strategies. The objective is to minimize technical discrepancies introduced by annotation mismatches or incompatible input formats.

Gene harmonization is treated as a key analytical component because previous studies have shown that deconvolution results are sensitive to reference composition, gene selection, and compatibility between bulk and single-cell modalities (Avila Cobos et al., 2020; Maden et al., 2023; Lu et al., 2025). All matching and filtering procedures are therefore documented explicitly within the workflow.

Stage 6: Implementation of Deconvolution Methods

Four deconvolution frameworks are considered in this thesis: EPIC, MuSiC, CIBERSORTx, and BisqueRNA. These methods were selected because they represent different but highly relevant approaches within the current state of the art of reference-based and scRNA-seq-guided deconvolution. EPIC is implemented through its R package and is particularly relevant for tumor samples because it was designed to estimate immune and cancer cell fractions while accounting for uncharacterized content (Racle et al., 2017). MuSiC is implemented in R and is specifically designed to use multi-subject single-cell references for bulk deconvolution (Wang et al., 2019). CIBERSORTx is included because it extends the original CIBERSORT framework

toward digital cytometry and broader RNA-seq applications (Newman et al., 2015, 2019). BisqueRNA is included as a single-cell-integrative method designed to account for systematic differences between bulk and single-cell data (Jew et al., 2020).

All methods are applied, as far as possible, under standardized preprocessing conditions and using harmonized input matrices in order to maximize comparability across approaches. Since these methods differ in implementation environment and input requirements, their outputs are interpreted comparatively rather than as if they were generated under perfectly identical assumptions.

Stage 7: Comparative and Robustness Evaluation

The comparative framework of this thesis focuses on methodological robustness rather than on direct clinical validation. The main analytical questions concern how strongly the outputs vary across methods and how sensitive they are to decisions made upstream in the workflow. Specifically, the comparative evaluation examines concordance of inferred cell proportions across methods, sensitivity to preprocessing choices, the impact of gene filtering and reference construction, and the extent to which outputs remain stable under a non-paired but biologically related reference setting.

Quantitative comparison includes cross-method correlation analyses, examination of variability patterns across analytical settings, and qualitative assessment of whether the inferred results remain biologically interpretable in light of the study context. In this sense, the comparison is not framed as a search for a universally best tool, but as an assessment of how different methods behave under a realistic and analytically constrained use case.

6.3 Reproducibility and Transparency

All analytical steps are implemented in R Markdown and version-controlled using Git to support reproducibility and transparent documentation of the workflow. The pipeline is modular, with clearly separated stages for preprocessing, annotation, reference construction, validation, harmonization, and deconvolution. This structure is intended to make methodological decisions traceable and to facilitate revision if parameters or preprocessing choices must be refined.

Reproducibility is treated as a core principle of the study rather than as a secondary technical feature. For this reason, filtering thresholds, normalization choices, annotation criteria, harmonization procedures, and deconvolution settings are documented explicitly. In addition, the methods section cites not only the deconvolution frameworks themselves, but also the data sources, preprocessing approaches, and methodological references that informed the analytical design. This is intended to ensure that the

workflow remains transparent, interpretable, and aligned with current standards for reproducible computational research.

7. Planning

7.1 Planning Strategy

The project follows an iterative and modular planning strategy aligned with the implemented analytical pipeline. Although the stages follow a logical dependency structure, the workflow allows refinement and adjustment of preprocessing parameters and reference construction decisions based on intermediate results.

The planning is organized around analytical milestones rather than isolated tasks, ensuring progressive validation and integration of results throughout the development process.

Estimated workload: 15–20 hours per week.

Phase 1 – Conceptual Framework and Literature Grounding

Timeline: Month 1

- Literature review on immune deconvolution methods.
- Definition of benchmarking criteria.
- Finalization of hypothesis and objectives.
- Selection and exploration of candidate datasets.

Milestone: Conceptual and methodological framework finalized.

Dependency: None.

Phase 2 – Single-Cell Processing and Reference Construction

Timeline: Months 1–2

- Quality control and normalization of scRNA-seq data.
- Clustering and marker identification.

- Semi-structured cell-type annotation.
- Construction of reference signature matrix.

Milestone: Final reference matrix generated and validated.

Dependency: Phase 1.

Potential risk:

- Ambiguous cluster annotation.
  Mitigation:
- Use of canonical markers and confidence-level filtering.

Phase 3 – Preliminary Technical Validation (Pseudobulk)

Timeline: Month 2

- Generation of pseudobulk mixtures.
- Implementation of selected deconvolution methods.
- Basic recovery assessment under controlled conditions.

Milestone: Technical validation completed.

Dependency: Phase 2.

Potential risk:

- Inconsistent method behavior due to parameter settings.
  Mitigation:
- Standardized input formatting and parameter documentation.

Phase 4 – Bulk RNA-seq Harmonization and Cross-Cohort Application

Timeline: Months 2–3

- Preprocessing and normalization of bulk RNA-seq data.

- Gene intersection and harmonization with reference matrix.
- Application of deconvolution methods to real bulk tumor samples.

Milestone: First complete cross-cohort deconvolution results.

Dependency: Phase 2 (reference matrix required).

Potential risk:

- Gene identifier mismatches.
  Mitigation:
- Explicit gene mapping and controlled filtering strategies.

Phase 5 – Comparative and Robustness Analysis

Timeline: Months 3–4

- Cross-method comparison.
- Sensitivity analysis to preprocessing and reference construction.
- Assessment of variability patterns under cross-cohort conditions.

Milestone: Robustness analysis completed.

Dependency: Phases 3 and 4.

Phase 6 – Writing, Integration, and Revision

Timeline: Months 3–5 (overlapping)

- Drafting of methods and results sections.
- Integration of figures and tables.
- Iterative revision based on supervisor feedback.

Milestone: Complete draft of thesis.

Dependency: Progressive completion of analytical stages.

7.3 Risk Assessment and Contingency Plan

The main methodological risks include:

- Non-matched dataset limitations.
- Sensitivity to reference construction choices.
- Computational inconsistencies across methods.
- Incomplete metadata availability.

These risks are mitigated through:

- Explicit acknowledgment of cross-cohort design.
- Preliminary pseudobulk validation.
- Standardized preprocessing procedures.
- Transparent documentation of assumptions and parameters.

7.4 Alignment with Academic Deliverables

The planning is aligned with the formal evaluation milestones (M1–M3):

- M1: Conceptual framework and methodological design.
- M2: Implementation and intermediate analytical results.
- M3: Final comparative evaluation and full thesis integration.

Gantt Chart

Each block ■ is roughly equivalent to one part of the month

Table 1 Gantt Chart

Task	M1	M2	M3	M4	M5	M6
Literature review & framework	■■	■■				

Table 1 Gantt Chart						
Dataset acquisition & exploration	■■					
scRNA preprocessing & clustering	■■	■■				
Reference construction		■■	■			
Pseudobulk validation		■■				
Bulk harmonization		■	■■			
Deconvolution methods implementation		■	■■	■		
Robustness & sensitivity analysis			■■	■■		
Interpretation & integration				■■	■	
Writing (progressive)	■	■■	■■	■■	■	
Revision & final adjustments					■■	■

The project begins with a conceptual and methodological grounding phase (M1), ensuring that benchmarking criteria and dataset selection are clearly defined before intensive computational analysis begins.

Single-cell preprocessing and reference construction (M1–M2) represent a foundational stage, as all subsequent analyses depend on the quality and reliability of the constructed reference matrix.

Preliminary pseudobulk validation (M2) functions as an internal methodological checkpoint before proceeding to real-world cross-cohort application.

Bulk harmonization and deconvolution implementation (M2–M3) form the core analytical phase of the project. Robustness and sensitivity analyses (M3–M4) extend this phase by evaluating methodological stability.

Writing activities overlap progressively with analytical stages to ensure early integration of results and reduce final-stage workload concentration.

Final revisions and integration (M5–M6) focus on consolidating the methodological narrative, discussing limitations, and ensuring coherence across sections.

8. State of art

8.1 Immune response in cancer: tumor microenvironment (TME)

Solid tumors are biologically complex tissues composed not only of malignant cells, but also of immune, stromal and vascular populations that together constitute the tumor microenvironment (TME). Interactions among these compartments influence tumor progression, metastatic behavior, immune evasion and treatment response. For this reason, the immune response in cancer cannot be understood only in terms of tumor-intrinsic biology, but must also be studied in relation to the cellular context in which malignant cells are embedded (Binnewies et al., 2018; Proietto et al., 2023).

Within this context, the distinction between hot and cold tumors is especially relevant. Hot tumors generally show higher immune infiltration, stronger antigen presentation and greater evidence of ongoing antitumor immune activity, whereas cold tumors tend to present limited immune infiltration or an immunosuppressive environment that constrains immune recognition and response. Although this distinction is a simplification, it is useful because it captures why studying immune populations in tumors may provide clinically and biologically relevant information, particularly in relation to prognosis, therapy response and resistance mechanisms (Galon & Bruni, 2019; Binnewies et al., 2018).

The role of immune cells in cancer is therefore not merely descriptive. Cytotoxic T lymphocytes, regulatory T cells, tumor-associated macrophages, dendritic cells, natural killer cells and cancer-associated fibroblasts can each contribute differently to disease evolution. Their abundance and functional state may alter the observed molecular profile of tumor tissue and shape the biological interpretation of transcriptomic data. Consequently, the characterization of tumor-associated immune populations has become an important objective in both basic and translational oncology research (Binnewies et al., 2018; Proietto et al., 2023).

8.2 Transcriptomic techniques to study the TME

Transcriptomic technologies provide complementary ways of studying the TME. Bulk RNA-seq remains the most widespread option for large tissue cohorts, whereas scRNA-seq offers cellular resolution and a more direct view of heterogeneity. Their strengths and limitations are therefore different rather than interchangeable.

8.2.1 Bulk RNA-seq

Bulk RNA sequencing measures the aggregate transcriptional signal of all cells present in a sample. Because of its relative maturity, robustness, lower cost and availability in public repositories, it has become one of the most commonly used technologies for transcriptomic profiling in both research and clinical studies. In oncology, bulk

RNA-seq has supported the study of tumor-intrinsic programs, differential expression, pathway activity and biomarker discovery across large cohorts (Newman et al., 2015; Newman et al., 2019).

At the same time, bulk RNA-seq has a central limitation in heterogeneous tissues: it does not preserve cellular resolution. As a consequence, the observed transcriptome reflects a mixture of malignant, immune and stromal signals. This means that differences in gene expression may arise either from regulation within a cell type or from changes in the abundance of that cell type in the tissue. In tumor samples, where immune infiltration and malignant content may vary strongly, this complicates biological interpretation and motivates the use of methods that can estimate cell-type composition from mixed transcriptomes (Avila Cobos et al., 2020; Maden et al., 2023).

8.2.2 scRNA-seq

Single-cell RNA sequencing was developed to address the loss of cellular resolution that characterizes bulk measurements. By profiling gene expression at the level of individual cells, scRNA-seq makes it possible to identify distinct cell populations, characterize cellular states, detect rare populations and study intratumoral heterogeneity more directly than bulk approaches (Luecken & Theis, 2019; Heumos et al., 2023).

In cancer research, these properties are especially valuable because the TME often contains diverse immune and stromal states whose transcriptional programs overlap or shift with disease stage and treatment context. Single-cell analyses have therefore become important for studying hot and cold tumors, therapy response, cellular resistance states and the structure of tumor-immune interactions. However, these advantages come at the cost of greater technical complexity, lower throughput per study compared with bulk cohorts, higher cost and stronger dependence on preprocessing, clustering and annotation decisions (Luecken et al., 2022; Nguyen et al., 2023).

Taken together, bulk RNA-seq and scRNA-seq should be viewed as complementary. Bulk remains attractive because it is cheaper and widely available for large cohorts, whereas scRNA-seq provides the cellular resolution that bulk lacks. It is in this gap between cost-effective tissue-level profiling and cell-resolved characterization that deconvolution methods become especially relevant. Their main strengths, limitations and typical applications in TME studies are summarized in Table 2

Table 2. Comparison of bulk RNA-seq and scRNA-seq for TME studies.

Technology	Main strengths	Main limitations	Typical use in TME studies
Bulk RNA-seq	Lower cost; large cohorts; mature workflows; tissue-level robustness	Mixed-cell signal; no direct cellular resolution	Large cohort profiling, differential expression, pathway

Table 2. Comparison of bulk RNA-seq and scRNA-seq for TME studies.

			analyses, downstream deconvolution
scRNA-seq	Cellular resolution; rare populations; state heterogeneity; reference construction	Higher cost; sparsity; dropout; stronger preprocessing dependence	Cell-type discovery, tumor ecosystem mapping, reference generation, pseudo-bulk benchmarking

8.3 Bulk deconvolution methods

Bulk deconvolution methods were developed to estimate cellular composition from mixed transcriptomic signals. In general, these approaches model the bulk profile as a combination of cell-type-specific reference signals and attempt to infer the relative contribution of each population. Their applications include immune infiltration estimation, tumor purity assessment and broader tissue composition analysis in both oncology and other biomedical settings (Newman et al., 2015; Racle et al., 2017; Im & Kim, 2023).

Benchmarking studies have shown that deconvolution performance depends not only on the algorithm, but also on the design of the reference, the transformation of the data, the definition of cell types and the biological scenario in which inference is performed. For this reason, current state-of-the-art discussions treat deconvolution as a pipeline-level methodological problem rather than as a simple software comparison (Avila Cobos et al., 2020; White et al., 2024; Xu et al., 2025; Nguyen et al., 2024).

8.3.1 Reference-free methods

Reference-free deconvolution methods aim to infer latent structure in the bulk data without requiring an external matrix of known cell-type signatures. These approaches can be useful when reliable reference profiles are unavailable or when the main objective is to identify hidden components rather than to quantify predefined cell types. However, the biological interpretation of inferred components may be less direct, and results can be sensitive to assumptions about matrix factorization and the number of latent components (Im & Kim, 2023; Nguyen et al., 2024).

Although the present thesis does not apply a reference-free algorithm, this family is relevant to the state of the art because it shows that deconvolution is not restricted to signature-based inference. It also highlights that the choice between reference-free and reference-based methods depends on data availability, biological interpretability and the intended use of the output.

8.3.2 scRNA-seq-guided reference-based methods

Reference-based deconvolution methods use known cell-type signatures to estimate the composition of bulk samples. Early approaches often relied on purified bulk populations and predefined marker sets. CIBERSORT established an influential machine-learning framework for robustly enumerating immune subsets from bulk tissue expression profiles (Newman et al., 2015), while later developments such as digital cytometry and CIBERSORTx extended these ideas to RNA-seq settings and richer abundance inference (Newman et al., 2019).

Other methods were developed with more specific tumor-oriented or single-cell-informed goals. EPIC was designed to estimate both cancer and immune cell fractions from bulk tumor data and explicitly handles uncharacterized content, which is especially relevant for solid tumors with incomplete reference coverage (Racle et al., 2017). MuSiC was specifically formulated to use multi-subject single-cell references and to exploit cross-subject consistency in cell-type-specific expression (Wang et al., 2019). Bisque addressed differences between bulk and single-cell measurements by introducing gene-specific transformations that improve the alignment of both modalities during proportion estimation (Jew et al., 2020).

The increasing use of scRNA-seq as a reference source has become one of the most important methodological transitions in bulk deconvolution. Single-cell datasets can provide cell-resolved profiles that capture greater biological heterogeneity than purified bulk signatures. However, this also means that reference construction depends on upstream choices such as quality control, normalization, clustering, annotation and the definition of cell-type granularity. In practice, scRNA-seq-guided deconvolution is therefore only as reliable as the compatibility between the reference and the bulk data, and as robust as the analytical decisions used to construct the reference itself (Luecken & Theis, 2019; Heumos et al., 2023; Xu et al., 2025). Table 3 summarizes the main analytical characteristics, strengths and limitations of the deconvolution methods considered in this thesis.

Table 3. Overview of representative deconvolution methods relevant to this thesis.

Method	Type	Reference requirement	Strengths	Main limitations / cautions
CIBERSORT / CIBERSORTx	Reference-based	Signature matrix from purified or scRNA-derived profiles	Widely used; strong immune focus; flexible framework	Sensitive to signature quality and compatibility
EPIC	Reference-based	Tumor-oriented reference with	Handles uncharacterized	Performance depends on

Table 3. Overview of representative deconvolution methods relevant to this thesis.

		immune and cancer content	content; suited to tumors	reference coverage
MuSiC	Reference-based	Multi-subject scRNA-seq reference	Leverages inter-subject consistency; natural scRNA integration	Affected by annotation granularity and dataset mismatch
BisqueRNA	Reference-based	scRNA-seq plus bulk integration	Accounts for bulk-scRNA shifts with gene-specific transformation	Requires careful preprocessing and matching
Reference-free methods	Unsupervised / latent	No external reference required	Useful when references are unavailable	Lower biological interpretability of latent components

Note. Own elaboration based on Newman et al. (2015), Newman et al. (2019), Racle et al. (2017), Wang et al. (2019), and Jew et al. (2020).

8.4 Bulk deconvolution applied in clinical and research scenarios

Deconvolution is now used in a wide range of research and applied contexts, from tumor immune profiling and tissue composition analysis to the interpretation of inflammatory or neurodegenerative signatures. In oncology, it is especially relevant because bulk cohorts remain abundant whereas matched single-cell measurements are much less common. This practical asymmetry explains why deconvolution often needs to operate under less than ideal reference conditions.

8.4.1 Studies performed with paired or non-paired data

In an ideal setting, bulk RNA-seq and scRNA-seq would be available from the same biological samples or at least from closely aligned specimens. Such paired data facilitate reference construction, reduce ambiguity during validation and provide a more favorable setting for assessing deconvolution performance. For this reason, many benchmarking studies use pseudo-bulk generated from single-cell data, experimentally mixed samples or datasets with orthogonal measurements such as flow cytometry (White et al., 2024; Huuki-Myers et al., 2025).

However, many real studies do not have direct one-to-one pairing between bulk and single-cell modalities. Instead, researchers must often use a biologically related but

non-matched single-cell dataset as the reference. This may happen because bulk and scRNA-seq were generated from different subsets of patients, from different tissue compartments or under different protocols. Recent work shows that such mismatch can affect deconvolution through missing cell types, altered transcriptome size, annotation differences and cross-platform incompatibility (Ivich et al., 2025; Lu et al., 2025; Maden et al., 2023).

These conditions do not invalidate deconvolution, but they do change the interpretation of benchmarking results. Under non-paired settings, robustness to preprocessing, gene harmonization and reference construction becomes at least as important as nominal algorithmic accuracy.

8.4.2 Context of application of this thesis: gaps and purpose

The literature clearly shows that bulk deconvolution is useful, that scRNA-seq references can substantially improve it and that no single method performs optimally across all tissues and evaluation frameworks. At the same time, several gaps remain. Many comparative studies still rely heavily on simulated mixtures or highly favorable reference conditions, and fewer works examine how commonly used methods behave when the available reference is biologically related but not directly matched to the bulk samples of interest (Tran et al., 2023; White et al., 2024; Xu et al., 2025).

The present thesis is positioned within this methodological gap. It does not propose a new deconvolution algorithm. Instead, it compares four widely used methods - CIBERSORTx, EPIC, MuSiC and BisqueRNA - under a realistic scenario in which the bulk and scRNA-seq datasets originate from the same broader study context but cannot be linked one-to-one at the sample level. This makes the study relevant not because it seeks a universally best method, but because it evaluates how analytical behavior changes when reference construction, preprocessing choices and gene harmonization become central determinants of the final output.

In that sense, the thesis contributes a context-specific assessment of methodological robustness in deconvolution. The expected contribution is practical guidance on how to interpret agreement and disagreement across methods when the underlying reference setting is biologically meaningful but analytically imperfect, which is a common situation in translational transcriptomics.

9. References

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